

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Objective:

1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

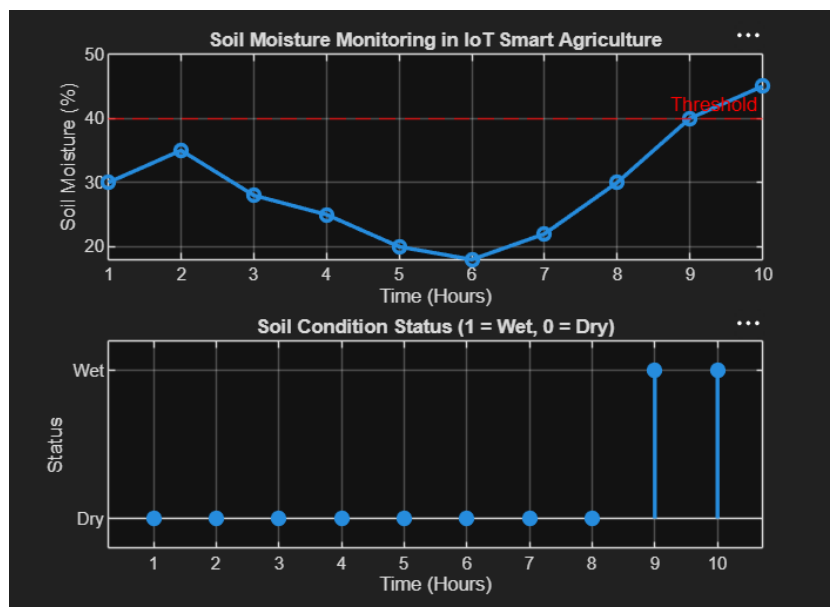
Objective 1 Matlab Code and Output:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
```

```

ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
output

```



Soil moisture levels vary over time, showing fluctuations in field water content. The threshold line helps identify dry and wet conditions for smart irrigation decisions.

Objective 2 Matlab Code and Output:

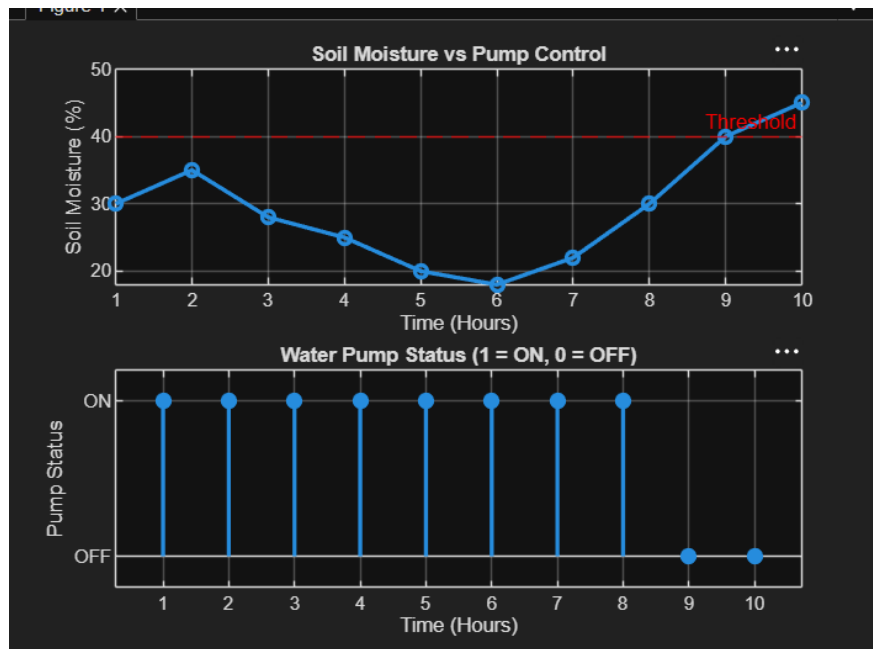
```
clc; clear; close all;
```

```

% Simulated soil moisture (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% ----- Graph 1: Soil Moisture vs Pump Status -----
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----
subplot(2,1,2)
stem(time, pump_status, 'filled','LineWidth',2)
title('Water Pump Status (1=ON, 0=OFF)')
xlabel('Time (Hours)')
ylabel('Pump Status')
grid on
code

```



The pump turns ON automatically when soil moisture drops below the threshold level. When moisture is sufficient, the pump remains OFF, ensuring efficient water usage and smart irrigation control.

Objective 3 Matlab Code and Output:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Irrigation efficiency calculation (%)
```

```
efficiency = (soil_moisture ./ threshold) * 100;
```

```
efficiency(efficiency > 100) = 100; % limit to  
100%
```

```
figure
```

```
% ----- Graph 1: Soil Moisture vs Efficiency
```

```
----- subplot(2,1,1)
```

```
plot(time, soil_moisture,
```

```
'-o','LineWidth',2) title('Soil Moisture  
Variation')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```
ylime(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Irrigation Efficiency
```

```
----- subplot(2,1,2)
```

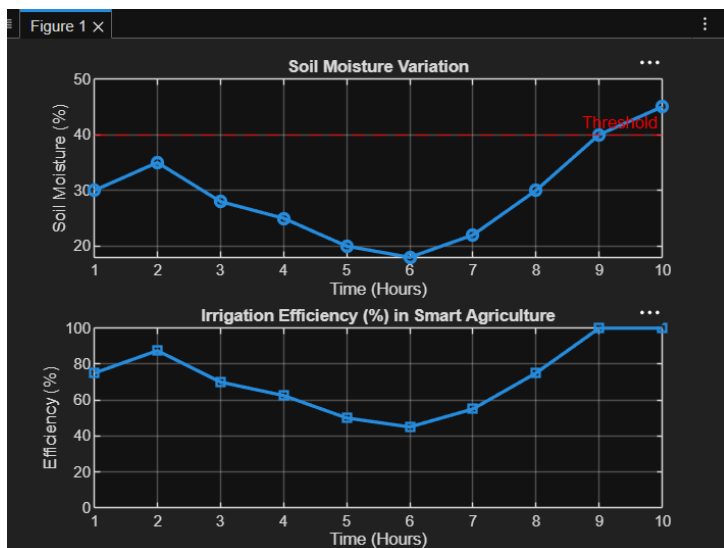
```
plot(time, efficiency, '-s','LineWidth',2)
```

```
title('Irrigation Efficiency (%) in Smart  
Agriculture') xlabel('Time (Hours)')
```

```
ylabel('Efficiency (%)')
```

```
grid on
```

```
Output
```



Irrigation efficiency increases when soil moisture approaches the optimal threshold level. The system ensures better water utilization by maintaining soil moisture within the desired range using smart control.

Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation,